

VLMs + MPC

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① Problem Formulation

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Consider the following nonlinear system:

$$\dot{x} = f(x, u)$$

where f is **unknown**. Our goal is to design a VLM-based controller $u = u_{vlm}(x, \mathcal{E})$ with formal guarantees, with $\mathcal{E}$ denoting environment variables.

Suppose that

- we have an MPC controller $u_{mpc}(x, \mathcal{E})$
- we have somehow learned a prediction model $\hat{f}(x, u)$
- we fine-tune $u_{vlm}(x, \mathcal{E})$ to imitate $u_{vlm}(x, \mathcal{E})$
- we have exchangeable N data samples of true trajectories x , nominal trajectories x_n , and environments $\mathcal{E}$ not used for learning, $\{(x_i, x_{n,i}, \mathcal{E}_i)\}_{i=1}^N$, where $\dot{x} = f(x, u_{vlm}(x, \mathcal{E}))$, $t \in [0, T]$ and $\dot{x}_n = \hat{f}(x_n, u_{mpc}(x_n, \mathcal{E}))$, $t \in [0, T]$
- we construct a contraction metric (<https://hirotsukamoto.com/contraction>) with $\hat{M}(x, t) \succ 0$ and $\underline{m}\mathbb{I} \preceq \hat{M} \preceq \overline{m}\mathbb{I}$ s.t.

$$\dot{\hat{M}} + \hat{M} \frac{\partial \hat{f}_{cl}}{\partial x} + \frac{\partial \hat{f}_{cl}}{\partial x} \hat{M} \preceq -2\alpha \hat{M} \quad (1)$$

is approximately true, where $\hat{f}_{cl}(x) = \hat{f}(x, u_{mpc}(x, \mathcal{E}))$ and $\alpha, \underline{m}, \overline{m} > 0$.

More specifically, for the Riemannian energy defined as

$$\hat{E}(\gamma(\cdot, t)) = \int_0^1 \delta_\gamma(s, t)^\top \hat{M}(\gamma(s, t)) \delta_\gamma(s, t) ds, \quad \delta_\gamma(s, t) = \frac{\partial \gamma(s, t)}{\partial s}$$

the contraction condition (1) implies that the following holds **approximately**:

$$\begin{aligned} \dot{\hat{E}}(\gamma(\cdot, t)) &= 2\delta_\gamma(1, t)^\top \hat{M}(\gamma(1, t)) f(x, u_{vlm}(x, \mathcal{E})) \\ &\quad - 2\delta_\gamma(0, t)^\top \hat{M}(\gamma(0, t)) \hat{f}(x_n, u_{mpc}(x_n, \mathcal{E})) \leq -2\alpha \hat{E}(\gamma(\cdot, t)) \end{aligned}$$

where γ is the minimizing geodesic connecting x and x_n [1] (**the derivative has to be the Dini derivative more precisely**), and x & x_n are again given by

$$\begin{aligned} \dot{x} &= f(x, u_{vlm}(x, \mathcal{E})) \\ \dot{x}_n &= \hat{f}(x_n, u_{mpc}(x_n, \mathcal{E}_n)). \end{aligned}$$

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How do we get formal guarantees in $\dot{x} = f(x, u_{vlm}(x, \mathcal{E}))$?

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Using conformal inference (<https://hirotsukamoto.com/contraction>), we get

$$\mathbb{P} \left[\left\| f(x, u_{vlm}(x, \mathcal{E})) - \hat{f}(x, u_{mpc}(x, \mathcal{E})) \right\| \leq q_{1-\delta}^{dyn}, \forall t \in [0, T] \right] \geq 1 - \delta^{dyn}$$

where

$$q_{1-\delta}^{dyn} = \lceil (1 - \delta^{dyn})(N + 1) \rceil\text{-th smallest of } (S_1^{dyn}, \dots, S_N^{dyn})$$

$$S_i^{dyn} = \sup_{t \in [0, T]} \left\| f(x, u_{vlm}(x_i, \mathcal{E}_i)) - \hat{f}(x, u_{mpc}(x_i, \mathcal{E}_i)) \right\|.$$

Using conformal inference once again, we also get

$$\mathbb{P}\left[2\delta_\gamma(1, t)^\top \hat{M}(\gamma(1, t))f(x, u_{vlm}(x, \mathcal{E})) - 2\delta_\gamma(0, t)^\top \hat{M}(\gamma(0, t))\hat{f}(x_n, u_{mpc}(x_n, \mathcal{E}_n)) \leq -2\alpha \hat{E}(\gamma(\cdot, t)) + q_{1-\delta}^{con}, \forall t \in [0, T]\right] \geq 1 - \delta^{con}$$

where

$$q_{1-\delta}^{con} = [(1 - \delta^{con})(N + 1)]\text{-th smallest of } (S_1^{con}, \dots, S_N^{con})$$

$$S_i^{con} = \max \left\{ 0, \sup_{t \in [0, T]} \left(2\delta_\gamma(1, t)^\top \hat{M}(\gamma(1, t))f(x, u_{vlm}(x, \mathcal{E})) - 2\delta_\gamma(0, t)^\top \hat{M}(\gamma(0, t))\hat{f}(x_n, u_{mpc}(x_n, \mathcal{E}_n)) + 2\alpha \hat{E}(\gamma(\cdot, t)) \right) \right\}.$$

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Let's consider the real-world closed-loop dynamical system controlled by $u_{vlm}(x, \mathcal{E})$ given as

$$\dot{x} = f(x, u_{vlm}(x, \mathcal{E})) = \hat{f}(x, u_{mpc}(x, \mathcal{E})) + \left(f(x, u_{vlm}(x, \mathcal{E})) - \hat{f}(x, u_{mpc}(x, \mathcal{E})) \right).$$

For our approximate Riemannian energy $\hat{E}(\gamma(\cdot, t))$, we get

$$\begin{aligned} \dot{\hat{E}}(\gamma(\cdot, t)) = & 2\delta_{\gamma}(1, t)^{\top} \hat{M}(\gamma(1, t)) f(x, u_{vlm}(x, \mathcal{E})) \\ & - 2\delta_{\gamma}(0, t)^{\top} \hat{M}(\gamma(0, t)) \hat{f}(x_n, u_{mpc}(x_n, \mathcal{E})) \\ & + 2\delta_{\gamma}(1, t)^{\top} \hat{M}(\gamma(1, t)) \left\| f(x, u_{vlm}(x, \mathcal{E})) - \hat{f}(x, u_{mpc}(x, \mathcal{E})) \right\|. \end{aligned} \quad (2)$$

It is known that the velocity field of a geodesic is parallel along the geodesic, which implies that $\hat{V} = \delta_\gamma(s, t)^\top \hat{M}(\gamma(s, t)) \delta_\gamma(s, t)$ remains constant in $s \in [0, 1]$ [2]. This gives

$$\hat{E}(\gamma(\cdot, t)) = \delta_\gamma(s, t)^\top \hat{M}(\gamma(s, t)) \delta_\gamma(s, t), \quad \forall s \in [0, 1]$$

and then

$$\hat{d}(x, x_n) = \sqrt{\hat{E}(\gamma(\cdot, t))} = \sqrt{\delta_\gamma(1, t)^\top \hat{M}(\gamma(1, t)) \delta_\gamma(1, t)} = \|\Theta(\gamma(1, t)) \delta_\gamma(1, t)\| \quad (3)$$

where $\hat{d}(x, x_n)$ is the Riemannian distance and Θ is given by $\hat{M} = \Theta^\top \Theta$.

Using (2) and (3), we get

$$\begin{aligned} \dot{\hat{E}}(\gamma(\cdot, t)) &= 2\delta_\gamma(1, t)^\top \hat{M}(\gamma(1, t))f(x, u_{vlm}(x, \mathcal{E})) \\ &\quad - 2\delta_\gamma(0, t)^\top \hat{M}(\gamma(0, t))\hat{f}(x_n, u_{mpc}(x_n, \mathcal{E})) \\ &\quad + 2\sqrt{m}\hat{d}(x, x_n) \left\| f(x, u_{vlm}(x, \mathcal{E})) - \hat{f}(x, u_{mpc}(x, \mathcal{E})) \right\|. \end{aligned}$$

Assuming that the test trajectories and environment $(x, x_n, \mathcal{E})$ in $t \in [0, T]$ are exchangeable with our data samples $\{(x_i, x_{n,i}, \mathcal{E}_i)\}_{i=1}^N$, we get

$$\begin{aligned} \mathbb{P} \left[\dot{\hat{E}}(\gamma(\cdot, t)) \leq -2\alpha \hat{E}(\gamma(\cdot, t)) + 2q_{1-\delta}^{dyn} \sqrt{m} \hat{d}(x, x_n) + q_{1-\delta}^{con}, \forall t \in [0, T] \right] \\ \geq 1 - \delta^{dyn} - \delta^{con} \end{aligned}$$

where [Boole's inequality](#) (a.k.a. the union bound) is also used here for getting the probability lower bound.

Using Young's inequality for products, i.e.,

$$2q_{1-\delta}^{dyn}\sqrt{\overline{m}}\hat{d}(x, x_n) \leq \varepsilon \hat{E}(\gamma(\cdot, t)) + \left(q_{1-\delta}^{dyn}\right)^2 \overline{m}\varepsilon^{-1}, \quad \forall \varepsilon > 0$$

we get

$$\begin{aligned} \mathbb{P} \left[\dot{\hat{E}}(\gamma(\cdot, t)) \leq -(2\alpha - \varepsilon)\hat{E}(\gamma(\cdot, t)) + \left(q_{1-\delta}^{dyn}\right)^2 \overline{m}\varepsilon^{-1} + q_{1-\delta}^{con}, \quad \forall t \in [0, T] \right] \\ \geq 1 - \delta^{dyn} - \delta^{con}. \end{aligned}$$

Finally, selecting ε to have $2\bar{\alpha} = 2\alpha - \varepsilon > 0$ yields

$$\mathbb{P} \left[\hat{E}(\gamma(\cdot, t)) \leq e^{-2\bar{\alpha}t} \hat{E}(\gamma(\cdot, 0)) + \frac{(q_{1-\delta}^{dyn})^2 \bar{m}\varepsilon^{-1} + q_{1-\delta}^{con}}{2\bar{\alpha}} (1 - e^{-2\bar{\alpha}t}), \forall t \in [0, T] \right] \\ \geq 1 - \delta^{dyn} - \delta^{con}.$$

- We could simplify this argument using Lyapunov
- Or use other performance metrics (e.g., constraint satisfaction, stability, recursive feasibility, as inspired by [3])
- We can use the bound for constraint tightening [1]

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- [2] M. Spivak. *A Comprehensive Introduction to Differential Geometry*. 3rd. Vol. 1. Wilmington, DE, USA: Publish or Perish, Inc., 1999.
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